

# **Gut Microbiota are Related to Parkinson's Disease and Clinical Phenotype**

**Scheperjans et al.**

## **Supplemental Data**

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## eMethods

### Study subjects

We aimed at minimizing the chance of recruiting patients with monogenic parkinsonism by restricting the age of motor-symptom onset to above 50 years and by excluding subjects with more than one relative or at least one first-degree relative with PD.<sup>1</sup> From the control group we furthermore excluded subjects with symptoms associated with premotor PD such as hyposmia, REM sleep behaviour disorder, and accumulation of other NMS (eTable 1).

Subjects were recruited from in- and outpatient departments of the participating hospitals as well as via referrals from cooperating neurologists. We also accepted referrals via participants and invited subjects via announcements in patient journals and meetings. Between November 2011 and December 2012, 277 subjects were screened for the study. Of these, 152 (76 patients, 76 controls) were included. All PD patients were under regular follow-up by a neurologist who was confident in the diagnosis. Patient records were reviewed if available. Thirty-six percent of patients were recruited by distinguished movement disorders specialists (SK, EK, JER). For the remainder, a local movement disorders specialist (JER, SK) confirmed diagnosis and approved inclusion if considered necessary by the investigator. Two subjects were excluded because no matching subject was available. Two subjects were excluded because acquisition of samples was delayed after closure of the analysis cohort. Four subjects were excluded because their stool samples yielded less than 4500 high-quality sequences.

### Clinical data

UPDRS assessments were done by FS (97%) and JER (3%). We classified the PD patients into postural instability and gait difficulty (PIGD), tremor dominant (TD), and mixed phenotypes (MX) as described by Jankovic et al.<sup>2</sup> In addition, we calculated an akinetic-rigid symptom score as described by Poletti et al., but we excluded items 29 (gait) and 30 (postural instability) since those are included in the PIGD score.<sup>3</sup>

For the detailed assessment of constipation there is no questionnaire available that is validated in PD patients. However, the Wexner score correlates well with objective physiologic findings in constipated patients and was therefore preferred over the Rome-III questionnaire which is designed for evaluation of functional bowel disorders.<sup>4, 5</sup> While diagnosis of active irritable bowel syndrome was an exclusion criterion in this study (eTable 1), we did not consider an isolated positive Rome-III questionnaire result an exclusion criterion since the specificity of the Rome criteria is limited.<sup>6</sup>

### Sample collection

The subjects collected the fecal samples at home into collection tubes pre-filled with Stool DNA Stabilizer (PSP® Spin Stool DNA Plus Kit, STRATEC Molecular). Within three days, the tubes were transferred to a storing temperature of -80 °C until further processing.

### DNA extraction and PCR

Total DNA was extracted using the PSP® Spin Stool DNA Plus Kit (STRATEC Molecular). PCR amplification was carried out in an ARKTIK Thermal Cycler (Finnzymes Diagnostics / Thermo Scientific). In the first stage, the V1–V3 regions of the bacterial 16S rRNA gene were amplified in three replicate reactions with universal bacterial primers pA (AGAGTTTGATCMTGGCTCAG) and pD' (GTATTACCGCGGCTGCTG) with 18-mer overhangs added to the 5' ends of the primers.<sup>7, 8</sup> The replicate PCR products were pooled and purified with Agencourt® AMPure® XP magnetic beads (Agencourt Bioscience) and subjected to a second PCR round with barcoded forward primers and a reverse primer, both of which attached to the respective 18-mer overhang sequences from the primers of the first PCR amplification. Phusion polymerase (Thermo Fisher Scientific / Finnzymes) with HF buffer and 2.5% DMSO were used. Cycling conditions for both PCR reactions consisted of an initial denaturation at 98 °C for 30 s, followed by 15 cycles at 98 °C for 10 s, 65 °C for 30 s, and 72 °C for 10 s, and then a final extension for 5 min. Between 3.6 and 60 ng of template DNA were used in the initial reaction. DNA concentration and quality were measured with Qubit (Invitrogen) and Bioanalyzer 2100 (Agilent).

## Sequencing and sequence quality control

PCR products were sequenced using 454-GS FLX Titanium chemistry (Roche Diagnostics). The dataset consisted of 2549217 raw reads (mean read count 17224 per subject). Basic sequence quality control and taxonomical assignment were performed with *mothur*, following the standard operating procedure for 454 sequenced 16S data.<sup>9-11</sup> The phylotype based approach was selected for further analysis. The final dataset included 1131504 reads, with a mean read count of 7645 per subject. The sequences have been deposited in the Sequence Read Archive at the European Bioinformatics Institute (accession no. PRJEB4927).

## Statistical analysis

Alpha diversity indices were calculated and beta diversity dendrograms describing the between subject differences were generated and compared with *mothur*.<sup>9-11</sup>

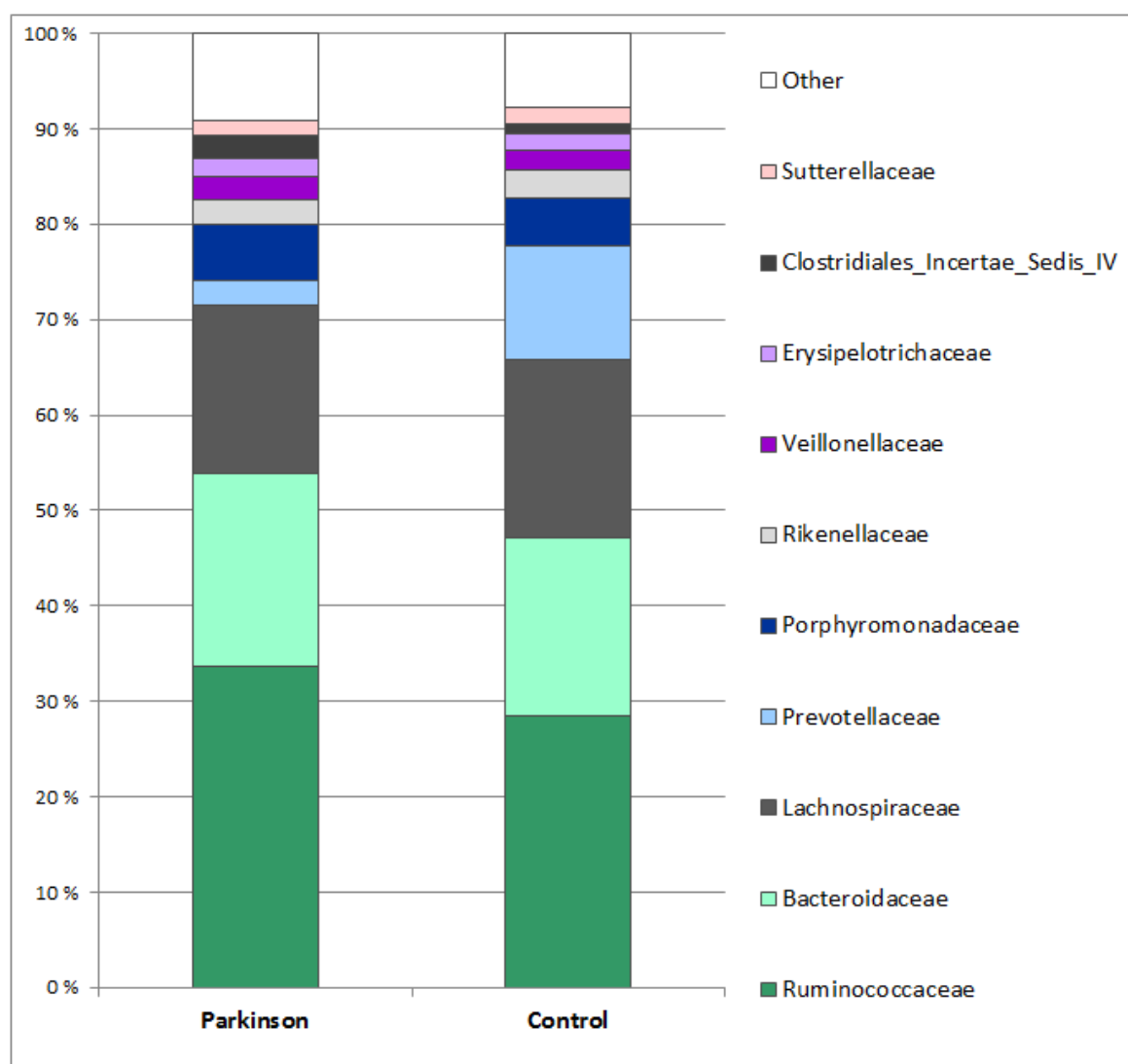
The following analyses were performed using IBM® SPSS® Statistics Version 19.0.0.1 (IBM Corp.): Group differences of clinical parameters were analyzed using t-tests for normally distributed variables, otherwise the Mann-Whitney U test was used. Differences regarding categorical variables were tested using Fisher's exact test. Associations of microbiome features with clinical parameters were explored using generalized linear models (GLM) employing linear, negative binomial, or logistic distributions depending on the distribution of the target variable.

In the Metastats analysis we analyzed those bacterial families that had more than 100 reads in the entire dataset. Differences with Q-values below 0.05 (1000 permutations) were considered significant and are the basis of the main conclusions of this paper. We also performed exploratory analyses on bacterial families with higher Q-values if P-values were below 0.05. In the remaining analyses (SPSS), P-values below 0.05 (two-sided if not indicated otherwise) were considered significant.

## eResults

### Microbiome

We investigated whether antiparkinsonian medication was a possible confounder by testing each of the families from eTable 4 for significantly different abundances between users and non-users of each antiparkinsonian drug class in the PD group. Patients using COMT-inhibitors had significantly higher levels (median % [IQR]) of Lactobacillaceae (0.22 [0-2.07] vs. 0 [0-0.02];  $p=0.001$ ) and lower levels of Clostridiales Incertae Sedis IV (0.00 [0-0.20] vs. 0.84 [0.02-2.73];  $p=0.005$ ) than those not using this class of drugs. No significant differences were found for any other antiparkinsonian drug class.

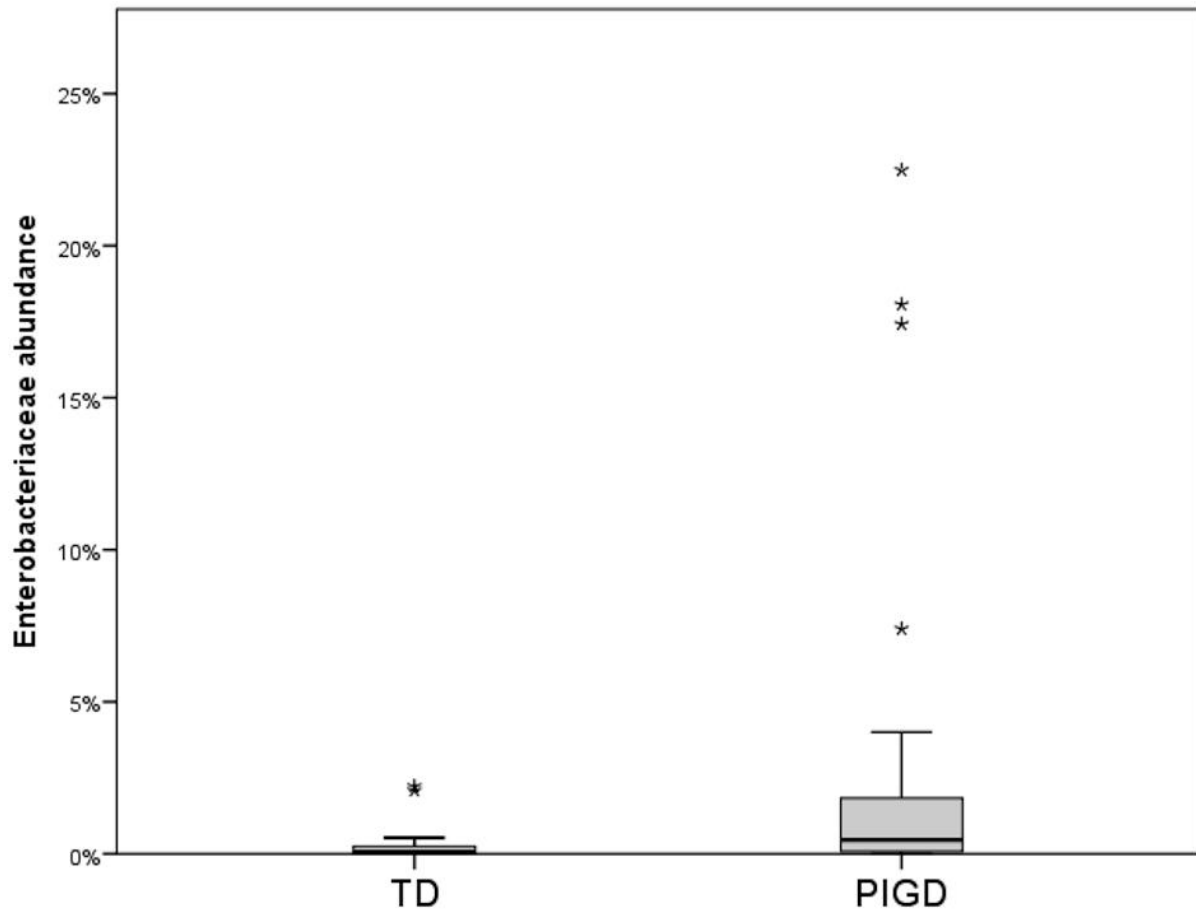
**eFigure 1****Ten most abundant bacterial families**

Relative abundances of the 10 most abundant bacterial families in the subsampled dataset.

The 10 and 5 most common families accounted for 91.6% and 81.3% of all reads, respectively.

**eFigure 2**

Distribution of Enterobacteriaceae in phenotypic subtypes of PD



Box plots showing the distributions of Enterobacteriaceae abundances in tremor dominant (TD) and postural instability and gait difficulty (PIGD) phenotypes. Black horizontal lines indicate the median values and the boxes around them delineate the IQR. Whiskers extend to the highest value within 1.5 IQR of the upper quartile. Asterisks represent extreme outliers beyond 3 IQR of the upper quartile. Median [IQR]: TD (0.04 [0.00-0.27]; n=23); PIGD (0.46 [0.07-1.84]; n=40)

**eTable 1**

## Inclusion and exclusion criteria

| Patients                                                                                             | Both groups                                                      | Controls                                                                                                            |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|                                                                                                      | <b>Inclusion criteria</b>                                        |                                                                                                                     |
|                                                                                                      | Written informed consent                                         |                                                                                                                     |
| Definite diagnosis of PD based on Queen Square brain bank clinical diagnostic criteria <sup>12</sup> |                                                                  |                                                                                                                     |
| Age of onset > 50 years                                                                              |                                                                  |                                                                                                                     |
|                                                                                                      | <b>Exclusion criteria</b>                                        |                                                                                                                     |
|                                                                                                      | Active smoking within last 6 months                              |                                                                                                                     |
|                                                                                                      | Diagnosis of dementia or MMSE < 25 points                        |                                                                                                                     |
|                                                                                                      | Diagnosis of major depression or GDS-15 > 9 points <sup>13</sup> |                                                                                                                     |
|                                                                                                      | Diagnosis of psychosis                                           |                                                                                                                     |
|                                                                                                      |                                                                  | Any signs of parkinsonism                                                                                           |
|                                                                                                      |                                                                  |                                                                                                                     |
|                                                                                                      |                                                                  | <u>Symptoms associated with premotor PD</u>                                                                         |
|                                                                                                      |                                                                  | Hyposmia (16-item sniffin sticks identification test result < 11 points) <sup>14</sup>                              |
|                                                                                                      |                                                                  | Rem sleep behaviour disorder (RBD diagnosis or RBDSQ > 4 points) <sup>15</sup>                                      |
|                                                                                                      |                                                                  | Restless legs syndrome (4-item RLS criteria) <sup>16</sup>                                                          |
|                                                                                                      |                                                                  | NMSQuest score > 3 points (items 4, 6, 7, 10, 18, 23, and 27 were not counted for screening purposes) <sup>17</sup> |



|  |                                                                                                                                                                                                                                                                                                                                                                |                                          |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
|  | <u>Environmental or genetic factors</u>                                                                                                                                                                                                                                                                                                                        |                                          |
|  | First degree relative or more than one relative with PD <sup>1</sup>                                                                                                                                                                                                                                                                                           |                                          |
|  | HIV infection                                                                                                                                                                                                                                                                                                                                                  |                                          |
|  |                                                                                                                                                                                                                                                                                                                                                                | Living in same household with PD patient |
|  |                                                                                                                                                                                                                                                                                                                                                                |                                          |
|  | <u>Factors related to or affecting gastrointestinal function</u>                                                                                                                                                                                                                                                                                               |                                          |
|  | Active or persistent primary disease of gastrointestinal tract: e.g. celiac disease, pernicious anemia, autoimmune gastritis, symptomatic diverticulosis, inflammatory bowel disease, irritable bowel syndrome, strictures, adhesions, varicosis or diverticulum of the esophagus, Meckel's diverticulum ( <b>Exception:</b> other forms of chronic gastritis) |                                          |
|  | Endocrinological disease ( <b>Exception:</b> diabetes mellitus without polyneuropathy; treated hypothyreosis with normal thyrotropin level)                                                                                                                                                                                                                    |                                          |
|  | Alcohol abuse                                                                                                                                                                                                                                                                                                                                                  |                                          |
|  | B-hypovitaminosis                                                                                                                                                                                                                                                                                                                                              |                                          |
|  | History of hepatobiliar or pancreatic disease ( <b>Exception:</b> asymptomatic cholecystolithiasis)                                                                                                                                                                                                                                                            |                                          |
|  | Previous abdominal or anorectal surgery ( <b>Exceptions:</b> can be enrolled after uncomplicated haemorrhoid procedure and 1 year after uncomplicated appendectomy, inguinal hernia repair, cholecystectomy, or gynecological surgery if symptoms resolved and no signs of adhesions)                                                                          |                                          |
|  | Severe gynaecological prolapse (grade III) ( <b>Exception:</b> can be enrolled after repair procedure if symptoms resolved and no signs of adhesions)                                                                                                                                                                                                          |                                          |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | History of peritonitis, severe endometriosis, polyneuropathy, polio, spina bifida, severe symptomatic spinal stenosis (if not symptom free for at least 1 year), paraparesis of any cause, symptomatic peripheral arteriosclerosis, any signs or history of intestinal ischemia (e.g. claudication), aortic aneurysm or dissection, connective tissue disease, autoimmune disease, sarcoidosis outside of the lungs or skin, cancer that is not considered cured, abdominal, intestinal, or urogenital fistula ( <b>Exceptions:</b> treated autoimmune hypothyreosis with normal thyrotropin level) |  |
|  | Heart failure ( <b>Exception:</b> isolated left ventricular heart failure NYHA ≤ II)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|  | Known severe renal insufficiency (glomerular filtration rate < 30 ml/min)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|  | One of the following within the previous 2 months: severe hypokalemia or hyperkalemia demanding hospital treatment, narcosis or analgosedation, endoscopic procedure of the gastrointestinal tract, abdominal trauma                                                                                                                                                                                                                                                                                                                                                                                |  |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|  | <u>Factors affecting gastrointestinal microbiota</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|  | Any of the following within the last 2 months: gastrointestinal or respiratory tract infection, food intoxication, major epistaxis requiring treatment by a physician                                                                                                                                                                                                                                                                                                                                                                                                                               |  |
|  | Antibiotic treatment within the last month <sup>18</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|  | <u>Drugs and medications</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|  | Any drug abuse                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |

|  |                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | Any regular use (>2 times a week) of the following medications over the last 2 months: opioids, loperamide, inhaled $\beta$ -agonists or anticholinergics, glucocorticoids (oral or parenteral), tricyclic antidepressants, antihistaminics with systemic anticholinergic effects, metoclopramide, cholinergics, anticholinergics (except for PD indication), domperidone, proton pump inhibitors |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**eTable 2**

Demographics, comorbidities, and medications

|                                                        | <b>Patients</b>  | <b>Controls</b>  | <b>P-value</b> |
|--------------------------------------------------------|------------------|------------------|----------------|
| <b>Demographics</b>                                    |                  |                  |                |
| n                                                      | 72               | 72               |                |
| female subjects                                        | 48.6%            | 50.0%            | 1              |
| Age (years, mean $\pm$ SD)                             | 65.3 $\pm$ 5.5   | 64.5 $\pm$ 6.9   | 0.448          |
| Body-mass-index (kg/m <sup>2</sup> , median [IQR])     | 26.3 [23.8-29.3] | 26.2 [23.7-28.1] | 0.795          |
| Relative with PD (max. one $\geq$ 2nd degree relative) | 20.0%            | 13.9%            | 0.376          |
|                                                        |                  |                  |                |
| <b>Comorbidities</b>                                   |                  |                  |                |
| Lactose Intolerance                                    | 12.5%            | 9.7%             | 0.792          |
| Diverticulosis (asymptomatic)                          | 1.4%             | 1.4%             | 1.000          |
| Irritable Bowel Syndrome (resolved)                    | 1.4%             | 2.8%             | 1.000          |
| Diabetes Mellitus Type 2                               | 11.1%            | 6.9%             | 0.562          |
| Hypothyroidism (controlled)                            | 11.1%            | 8.3%             | 0.780          |
| Hyperthyroidism (resolved)                             | 4.2%             | 0 %              | 0.245          |
| Appendectomy                                           | 11.1%            | 9.7%             | 1.000          |
| Cholecystectomy                                        | 12.5%            | 4.2%             | 0.129          |
| Inguinal Hernia Repair                                 | 6.9%             | 4.2%             | 0.719          |
| Other Gastrointestinal Tract Surgery                   | 15.3%            | 8.3%             | 0.302          |
| Gynaecological Prolapse                                | 5.7%             | 8.3%             | 1.000          |
| Gynaecological Prolapse Repair                         | 5.7%             | 5.6%             | 1.000          |

|                                                                              |       |       |         |
|------------------------------------------------------------------------------|-------|-------|---------|
| Endometriosis                                                                | 2.9%  | 5.6%  | 1.000   |
| Spinal Stenosis<br>(currently asymptomatic)                                  | 2.8%  | 2.8%  | 1.000   |
| Peripheral arterial<br>occlusive disease                                     | 1.4%  | 0 %   | 1.000   |
| Autoimmune Disease                                                           | 4.3%  | 2.8%  | 0.679   |
| Sarcoidosis                                                                  | 0 %   | 0 %   |         |
| Cancer                                                                       | 5.6%  | 4.2%  | 1.000   |
| Heart Failure                                                                | 1.4%  | 0 %   | 1.000   |
| Arterial Hypertension                                                        | 36.1% | 45.8% | 0.309   |
| Coronary Artery Disease                                                      | 8.5%  | 9.7%  | 1.000   |
| Myocardial Infarction                                                        | 0 %   | 4.2%  | 0.245   |
| Atrial Fibrillation                                                          | 4.2%  | 18.1% | 0.015   |
| Hypercholesterolaemia<br>(LDL > 3 mmol/l or total<br>cholesterol > 5 mmol/l) | 70.7% | 85.2% | 0.075   |
| TIA or Ischemic Stroke                                                       | 7.0%  | 37.5% | < 0.001 |
| Intracranial<br>Haemorrhage                                                  | 0 %   | 2.8%  | 0.497   |
| Subarachnoid<br>Haemorrhage                                                  | 0 %   | 0 %   |         |
|                                                                              |       |       |         |
| <b>Medication</b>                                                            |       |       |         |
| Levodopa                                                                     | 54.2% | 0 %   | < 0.001 |
| COMT Inhibitor                                                               | 15.3% | 0 %   | 0.001   |
| Dopamine Agonist                                                             | 77.8% | 0 %   | < 0.001 |
| MAO Inhibitor                                                                | 70.8% | 0 %   | < 0.001 |
| Anticholinergic                                                              | 8.3%  | 0 %   | 0.028   |
| Amantadine                                                                   | 2.8%  | 0 %   | 0.497   |
| Betablocker                                                                  | 22.2% | 29.2% | 0.446   |
| Selective Serotonin<br>Reuptake Inhibitor                                    | 8.3%  | 2.8%  | 0.275   |

|                                  |       |       |         |
|----------------------------------|-------|-------|---------|
| Acetylsalicylic Acid             | 22.2% | 26.4% | 0.698   |
| Warfarin                         | 1.4%  | 15.3% | 0.004   |
| ACE-inhibitor or ATR-antagonist  | 30.6% | 45.8% | 0.086   |
| Diuretic                         | 13.9% | 11.1% | 0.802   |
| Calcium Channel Blocker          | 6.9%  | 15.3% | 0.184   |
| Thyroxine                        | 11.1% | 8.3%  | 0.780   |
| Antiepileptic                    | 0 %   | 0 %   |         |
| Statin                           | 20.8% | 54.2% | < 0.001 |
| Laxatives                        | 2.8%  | 0 %   | 0.497   |
| Probiotics                       | 29.2% | 23.6% | 0.571   |
| Opioids*                         | 1.4%  | 2.8%  | 1.000   |
| Drugs for Treatment of Diarrhea* | 1.4%  | 0 %   | 1.000   |
| Cortisone*                       | 2.8%  | 5.6%  | 0.681   |
| Tricyclic Antidepressants*       | 0 %   | 0 %   |         |
| Antihistaminics*                 | 1.4%  | 2.8%  | 1.000   |
| Antiemetics*                     | 0 %   | 0 %   |         |
| Anacidics*                       | 8.3%  | 12.5% | 0.587   |

\* = Twice a week or less within preceding 2 months, otherwise exclusion criterion

**eTable 3**

Clinical characteristics of the PD cohort

| <b>H&amp;Y Stage</b>                                      | <b>1</b>            | <b>1.5</b>      | <b>2</b>        | <b>2.5</b>      | <b>3</b>   |
|-----------------------------------------------------------|---------------------|-----------------|-----------------|-----------------|------------|
| n (%)                                                     | 4 (5.6%)            | 2 (2.8%)        | 23 (31.9%)      | 27 (37.5%)      | 16 (22.2%) |
|                                                           |                     |                 |                 |                 |            |
|                                                           | <b>All patients</b> | <b>TD</b>       | <b>MX</b>       | <b>PIGD</b>     |            |
| n (%)                                                     | 72 (100)            | 23 (31.9)       | 9 (12.5)        | 40 (55.6)       |            |
| UPDRS-III<br>mean $\pm$ SD                                | 31.6 $\pm$ 9.0      | 32.1 $\pm$ 8.6  | 24.8 $\pm$ 9.9  | 32.8 $\pm$ 8.5  |            |
| UPDRS total<br>(I-IV)<br>mean $\pm$ SD                    | 45.5 $\pm$ 13.6     | 42.9 $\pm$ 11.2 | 36.9 $\pm$ 15.2 | 49.0 $\pm$ 13.7 |            |
| Years from<br>motor onset;<br>median [IQR];<br>n=71       | 5 [3-8]             | 5 [3-6]         | 6 [2-12.5]      | 6 [3-10]        |            |
| Years from<br>NMS onset;<br>median [IQR];<br>n=58         | 6.5 [2.75-11]       | 6 [4-10]        | 4 [1-10]        | 7 [2-14]        |            |
| Dopamine<br>transporter<br>SPECT<br>documented*;<br>n (%) | 26 (36.1%)          | 13 (56.5%)      | 2 (22.2%)       | 11 (27.5%)      |            |

TD = tremor dominant phenotype; MX = mixed phenotype; PIGD = postural instability and gait difficulty phenotype <sup>2</sup>

\* For 14 (19.4%) patients it was not known whether a SPECT study had been performed at time of diagnosis.

**eTable 4**

Metastats contrast PD vs. Control

| <b>Family</b>                   | <b>Patients</b> | <b>Controls</b> | <b>P-value</b> | <b>Q-value</b> |
|---------------------------------|-----------------|-----------------|----------------|----------------|
| Prevotellaceae                  | 2.70 ± 0.32     | 12.06 ± 3.73    | 0.001          | 0.031          |
| Lactobacillaceae                | 0.44 ± 0.04     | 0.02 ± 0.00     | 0.004          | 0.063          |
| Verrucomicrobiaceae             | 0.06 ± 0.00     | 0.02 ± 0.00     | 0.014          | 0.146          |
| Bradyrhizobiaceae               | 0.16 ± 0.00     | 0.03 ± 0.00     | 0.021          | 0.151          |
| Clostridiales Incertae Sedis IV | 2.49 ± 0.30     | 1.01 ± 0.03     | 0.025          | 0.151          |
| Ruminococcaceae                 | 33.63 ± 1.99    | 28.54 ± 1.81    | 0.029          | 0.151          |

Bacterial families showing different abundances (mean % ± variance) between PD and control groups with a P-value less than 0.05.



**eTable 5**

GLM results (Control vs. PD; IBS excluded)

| Family                          | Control vs. PD                                 | Atrial fibrillation No vs. Yes                  | TIA or ischemic stroke No vs. Yes              | Warfarin No vs. Yes                            | Statin No vs. Yes                              | Total NMSS Score                               | Total Wexner Score                           | Prevotellaceae abundance                             | COMT-inhibitor No vs. Yes                      |
|---------------------------------|------------------------------------------------|-------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|----------------------------------------------|------------------------------------------------------|------------------------------------------------|
| Prevotellaceae                  | 1.469<br>[0.955-1.982],<br>31.397, <0.001      | 1.343<br>[0.227-2.460],<br>5.558, 0.018         | -0.177<br>[-0.747-0.394],<br>0.368, 0.544      | -1.013<br>[-2.234-0.208],<br>2.646, 0.104      | -0.262<br>[-0.757-0.233],<br>1.077, 0.299      | 0.047<br>[-0.195-0.288],<br>0.144, 0.704       | 0.306<br>[-0.016-0.628],<br>3.471, 0.062     |                                                      |                                                |
| Lactobacillaceae                | -4.083<br>[-5.367 - -2.798],<br>38.790, <0.001 | -0.475<br>[-2.105-1.155],<br>0.326, 0.568       | 2.070<br>[1.175-2.965],<br>20.554, <0.001      | 0.040<br>[-1.728-1.809],<br>0.002, 0.964       | -1.906<br>[-2.529 - -1.284],<br>36.018, <0.001 | 0.278<br>[-0.251-0.808],<br>1.062, 0.303       | 0.017<br>[-0.358-0.392],<br>0.008, 0.928     | 0.382<br>[0.139-0.624],<br>9.526, 0.002              | -3.577<br>[-4.523 - -2.631],<br>54.952, <0.001 |
| Verrucomicrobiaceae             | -1.138<br>[-1.959 - -0.317],<br>7.380, 0.007   | 0.628<br>[-1.201-2.457],<br>0.453, 0.501        | 0.470<br>[-0.430-1.370],<br>1.048, 0.306       | -1.568<br>[-3.522-0.387],<br>2.471, 0.116      | 0.221<br>[-0.493-0.934],<br>0.367, 0.544       | -0.370<br>[-0.919-0.179],<br>1.744, 0.187      | 0.099<br>[-0.343-0.542],<br>0.194, 0.660     | -0.928<br>[-1.518 - -0.338],<br>9.494, 0.002         |                                                |
| Bradyrhizobiaceae               | -2.052<br>[-2.821 - -1.283],<br>27.376, <0.001 | 0.089<br>[-1.235-1.412],<br>0.017, 0.896        | -1.172<br>[-2.112 - -0.232],<br>5.971, 0.015   | 2.927<br>[0.464-5.389],<br>5.423, 0.020        | 1.386<br>[0.599-2.173],<br>11.913, 0.001       | -0.617<br>[-1.076 - -0.158],<br>6.948, 0.008   | -0.470<br>[-0.869 - -0.071],<br>5.334, 0.021 | -0.266<br>[-0.615-0.083],<br>2.239, 0.135            |                                                |
| Clostridiales Incertae Sedis IV | 1.803<br>[0.838 - 2.769],<br>13.400, <0.001    | 0.172<br>[-0.501-0.845],<br>0.252, 0.616        | 0.176<br>[-0.347-0.698],<br>0.435, 0.510       | 0.589<br>[-0.154-1.332],<br>2.414, 0.120       | -1.158<br>[-1.674 - -0.643],<br>19.395, <0.001 | 0.144<br>[-0.223-0.512],<br>0.593, 0.441       | 0.211<br>[-0.131-0.553],<br>1.459, 0.227     | -0.795<br>[-1.024 - -0.566],<br>46.248, <0.001       | 2.136<br>[1.303-2.969],<br>25.270, <0.001      |
| Ruminococcaceae                 | -51.247<br>[-346.199-243.705],<br>0.116, 0.733 | -325.658<br>[-853.213-201.897],<br>1.464, 0.226 | 138.206<br>[-154.680-431.091],<br>0.855, 0.355 | 202.028<br>[-373.196-777.252],<br>0.474, 0.491 | -207.582<br>[-454.753-39.590],<br>2.709, 0.100 | -21.118<br>[-198.513-156.278],<br>0.054, 0.816 | 152.442<br>[0.128-304.755],<br>3.848, 0.050  | -212.289<br>[-314.754 - -109.825],<br>16.489, <0.001 |                                                |

Results of the GLMs for bacterial abundances (sequence counts) based on the group factor and possible confounders. For this calculation, subjects fulfilling IBS criteria based on the Rome-III questionnaire were excluded (n=122). Normal distribution for Ruminococcaceae. Negative binomial distribution with log link for all others. COMT-inhibitor medication effect was nested within the study group effect for Lactobacillaceae and Clostridiales Incertae Sedis IV. Based on z-transformed values. Results shown as: B [95% CI], Wald Chi-Square, P-value

**eTable 6**

Logistic Regression PD vs. Control (Bacteria only)

| <b>Covariate</b>                | <b>Wald</b> | <b>P-value</b> | <b>OR per 1% increase in abundance for belonging to PD group [95% CI]</b> |
|---------------------------------|-------------|----------------|---------------------------------------------------------------------------|
| Prevotellaceae                  | 7.478       | 0.006          | 0.945<br>[0.907-0.984]                                                    |
| Lactobacillaceae                | 1.805       | 0.179          | 23.441<br>[0.235-2337.377]                                                |
| Bradyrhizobiaceae               | 3.965       | 0.046          | 15.788<br>[1.044-238.716]                                                 |
| Clostridiales Incertae Sedis IV | 2.166       | 0.141          | 1.112<br>[0.966-1.280]                                                    |

Results of logistic regression analysis with study group as the dependent variable. Initially all six bacterial families were included as covariates. We employed a backward elimination method based on the likelihood ratio.

**eTable 7**

Logistic Regression PD vs. Control (Bacteria and Constipation Score)

| <b>Covariate</b>                | <b>Wald</b> | <b>P-value</b> | <b>OR per 1 unit z-score increase for belonging to PD group [95% CI]</b> |
|---------------------------------|-------------|----------------|--------------------------------------------------------------------------|
| Prevotellaceae                  | 7.053       | 0.008          | 0.429<br>[0.230-0.801]                                                   |
| Lactobacillaceae                | 1.752       | 0.186          | 156.372<br>[0.088-277179.530]                                            |
| Bradyrhizobiaceae               | 3.562       | 0.059          | 2.590<br>[0.964-6.957]                                                   |
| Clostridiales Incertae Sedis IV | 1.084       | 0.298          | 1.349<br>[0.768-2.371]                                                   |
| Wexner total score              | 17.742      | <0.001         | 2.969<br>[1.789-4.927]                                                   |

Results of logistic regression analysis with study group as the dependent variable after adding Wexner constipation score to the model. Based on z-transformed values.

**eTable 8**

GLM results (UPDRS-III)

| <b>Family</b>                      | <b>UPDRS-III total score<br/>(all patients included; n=72)</b> | <b>UPDRS-III total score<br/>(motor fluctuations and DBS<br/>excluded; n=43)</b> |
|------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------|
| Prevotellaceae                     | 0.420 [0.067-0.772], 5.452, 0.020                              | 0.383 [-0.060-0.826], 2.867, 0.090                                               |
| Lactobacillaceae                   | 0.552 [-0.570-1.673], 0.929, 0.335                             | 1.396 [-6.041-8.832], 0.135, 0.713                                               |
| Verrucomicrobiaceae                | -5.967 [-19.582-7.648], 0.738, 0.390                           | -20.142 [-53.934-13.651], 1.365, 0.243                                           |
| Bradyrhizobiaceae                  | 3.826 [-0.111 – 7.762], 3.629, 0.057                           | 4.116 [-0.894-9.125], 2.593, 0.107                                               |
| Clostridiales Incertae<br>Sedis IV | 0.273 [-0.087-0.632], 2.205, 0.138                             | 0.301 [-0.115-0.717], 2.007, 0.157                                               |

Results of the GLMs modelling UPDRS-III total score using abundances (%) of Prevotellaceae and those four families that had shown significant study group effects in the comparison between PD patients and controls (Table 2). Normal distribution. Results shown as: B [95% CI], Wald Chi-Square, P-value

**eTable 9**

Metastats contrast TD vs. PIGD

| <b>Family</b>       | <b>TD</b>   | <b>PIGD</b> | <b>P-value</b> | <b>Q-value</b> |
|---------------------|-------------|-------------|----------------|----------------|
| Enterobacteriaceae  | 0.28 ± 0.00 | 2.31 ± 0.26 | 0.004          | 0.018          |
| Erysipelotrichaceae | 1.17 ± 0.02 | 2.19 ± 0.04 | 0.024          | 0.103          |

Bacterial families showing differences in abundances (mean % ± variance) between TD and PIGD phenotypes with a P-value less than 0.05.

**eTable 10**

Metastats contrast TIA/ischemic stroke

| <b>Family</b>       | <b>Previous<br/>TIA/ischemic<br/>stroke</b> | <b>NO previous<br/>TIA/ischemic<br/>stroke</b> | <b>P-value</b> | <b>Q-value</b> |
|---------------------|---------------------------------------------|------------------------------------------------|----------------|----------------|
| Cytophagaceae       | 0.00 ± 0.00                                 | 0.03 ± 0.00                                    | < 0.001        | 0.021          |
| Succinivibrionaceae | 0.00 ± 0.00                                 | 0.06 ± 0.00                                    | < 0.001        | 0.021          |
| Coriobacteriaceae   | 0.14 ± 0.00                                 | 0.29 ± 0.00                                    | 0.002          | 0.029          |
| Clostridiaceae_2    | 0.01 ± 0.00                                 | 0.02 ± 0.00                                    | 0.007          | 0.075          |
| Lactobacillaceae    | 0.02 ± 0.00                                 | 0.28 ± 0.03                                    | 0.038          | 0.326          |

Bacterial families showing different abundances (mean % ± variance) between subjects with and without previous TIA/ischemic stroke with a P-value less than 0.05.

**eTable 11**

Metastats contrast Atrial Fibrillation

| <b>Family</b>                  | <b>Atrial<br/>fibrillation</b> | <b>NO atrial<br/>fibrillation</b> | <b>P-value</b> | <b>Q-value</b> |
|--------------------------------|--------------------------------|-----------------------------------|----------------|----------------|
| Cytophagaceae                  | 0.00 ± 0.00                    | 0.02 ± 0.00                       | < 0.001        | 0.016          |
| Marinilabiaceae                | 0.00 ± 0.00                    | 0.03 ± 0.00                       | < 0.001        | 0.016          |
| Burkholderiales Incertae Sedis | 0.00 ± 0.00                    | 0.05 ± 0.00                       | < 0.001        | 0.016          |
| Succinivibrionaceae            | 0.00 ± 0.00                    | 0.05 ± 0.00                       | 0.002          | 0.024          |

Bacterial families showing different abundances (mean % ± variance) between subjects with and without atrial fibrillation with a P-value less than 0.05.

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